

R-MDP/POMDP

1 Abstract

2 Introduction

add an illustrative figure over the theory approach

3 Preliminaries

We consider a finite-horizon partially observable Markov decision process (POMDP) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{Z}, P_T, P_Z, r, \mathcal{H})$, where $\mathcal{S}, \mathcal{A}$, and $\mathcal{Z}$ denote the finite state, action, and observation spaces, respectively. The transition model $P_T(s' | s, a)$ defines the probability of transitioning from state $s \in \mathcal{S}$ to state $s' \in \mathcal{S}$ after taking action $a \in \mathcal{A}$. The observation model $P_Z(z | s')$ defines the probability of receiving observation $z \in \mathcal{Z}$ after arriving at state s' . The reward function $r : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is assumed to be bounded, with $|r(s, a)| \leq R_{\max} \forall s \in \mathcal{S}, a \in \mathcal{A}$. The horizon is denoted by $\mathcal{H}$. **should i also introduce MDP??**

At each time step t the agent does not observe the underlying state directly, and therefore decisions cannot be conditioned on s_t . Instead, the information available to the agent is summarized by the history of actions and observations up to time t which we denote by $h_t = \{a_0, z_1, \dots, a_{t-1}, z_t\}$. This history induces a posterior distribution over the current hidden state, called the belief, defined by $b_t(s) = \Pr(s_t = s | h_t)$. at each time step t after taking action a_t and observing z_{t+1} , the belief is updated with the bayesian filter $b_{t+1}(s') = \Pr(s_{t+1} = s' | h_t, a_t, z_{t+1}) = \eta_t \cdot P_Z(z_{t+1} | s') \sum_{s \in \mathcal{S}} P_T(s' | s, a_t) b_t(s)$, where η_t is the normalizing constant.

A policy specifies how actions are selected from the information available to the agent. Since the agent observes histories rather than states, we consider history-dependent policies **(we are not really limited to history dependent policies, should discuss what should be presented)**. For a finite-horizon problem, a policy is a sequence $\pi = \{\pi_t\}_{t=0}^H$, where each π_t maps a history h_t to a distribution over actions: $\pi_t(a | h_t) = \Pr(a_t = a | h_t)$, $a \in \mathcal{A}$. Thus, under a stochastic policy, the action a_t is sampled according to $\pi_t(\cdot | h_t)$. Deterministic policies are obtained as a special case, where all probability mass is assigned to a single action; in that case we write $a_t = \pi_t(h_t)$. The analysis below is stated for history-dependent policies, and therefore applies to both stochastic policies and deterministic tree policies.

Given a policy π , the value function measures the expected cumulative reward obtained from the information available at the current history. Starting from history h_t , the future evolution is random because the hidden state is distributed according to the belief b_t , the system evolves according to the transition and observation models, and, in the stochastic case, actions are sampled from the policy. For a planning horizon H , we define the return from time t as:

$$G_t^\pi = \sum_{j=t}^{t+H} r(s_j, a_j), \quad a_j \sim \pi_j(\cdot | h_j) \quad (1)$$

The value of policy π at history h_t is then the conditional expectation of this return:

$$V_t^\pi(h_t) = \mathbb{E}^\pi \left[\sum_{j=t}^{t+H} r(s_j, a_j) \middle| h_t, a_j \sim \pi_j(\cdot | h_j) \right] = \quad (2)$$

$$\sum_{j=t}^{t+H} \sum_{z_{t+1:j}} \sum_{s_{t:j}} \prod_{k=t}^{j-1} P_T(s_{k+1} | \pi_k, s_k) P_Z(z_{k+1} | s_{k+1}) b(s_t | h_t) r(s_j, a_j) \quad (3)$$

the derivation of 3 is in the supplementary

The expectation is taken over the current hidden state $s_t \sim b_t$, the future state-observation sequence generated by P_T and P_Z , and the future actions generated by π . In the deterministic case, where $a_j = \pi_j(h_j)$, the same definition applies with no additional expectation over actions. This value function is the full-space quantity that the online planner seeks to approximate.

4 Uncertainty Set

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our online planner (or policy evaluator) is robust with respect to **sampled trajectories** during planning, which implies that the uncertainty sets significantly differs from the theoretical uncertainty set. let us first define the theoretical uncertainty set and from that define the uncertainty set of the simplified planner. given some nominal models P_T^o, P_Z^o and some distance metric $D(\cdot || \cdot)$ which defines the uncertainty in the models (it can be different uncertainty for each model $D_T(\cdot || \cdot), D_Z(\cdot || \cdot)$ we define the uncertainty sets over the models as:

$$\mathcal{P}_T(s, a) = \{P_T(\cdot | s, a) : D_T(P_T(\cdot | s, a) || P_T^o(\cdot | s, a)) \leq \rho(s, a)\} \quad (4)$$

$$\mathcal{P}_Z(s) = \{P_Z(\cdot | s) : D_Z(P_Z(\cdot | s') || P_Z^o(\cdot | s')) \leq \rho(s)\} \quad (5)$$

for every $(s, a)/(s)$ we get a different set, with possibly different radius $(\rho(s, a)/\rho(s))$. this is the most general definition for the uncertainty, and does not require us to assume further assumptions.

we introduce a new concept we call "the projected uncertainty set". during online planning the planner is exposed to a subset of the true world, and thus we can only plan according to samples we have seen. the projected uncertainty set is defined as sets of valid sub-probability vectors such that each sub-vector can be extended into a full valid distribution that lies within the original uncertainty set, and therefore we are guaranteed (maybe we should add some more formal/mathematical derivation that justify this guarentee) that the values we compute represent the true worst case of the current policy with respect to the problem error radius. the projected sets are thus:

$$\hat{\mathcal{P}}_T(s, a; h) = \left\{ P_T^{\text{in}} \in \mathbb{R}_+^{|S^{\text{in}}|} : \exists P_T^{\text{out}} \in \mathbb{R}_+^{|S^{\text{out}}|} \text{ s.t. } (P_T^{\text{in}}, P_T^{\text{out}}) \in \mathcal{P}_T \right\} \quad (6)$$

$$\hat{\mathcal{P}}_Z(s; ha) = \left\{ P_Z^{\text{in}} \in \mathbb{R}_+^{|Z^{\text{in}}|} : \exists P_Z^{\text{out}} \in \mathbb{R}_+^{|Z^{\text{out}}|} \text{ s.t. } (P_Z^{\text{in}}, P_Z^{\text{out}}) \in \mathcal{P}_Z \right\} \quad (7)$$

the history node h stores the sampled states, and the action node ha stores the observations seen so far. more specifically:

$$h = \{s_i\}_{i=0}^{N(h)}, \quad ha = \{z_i\}_{i=0}^{N(ha)} \quad (8)$$

although 8 allows duplicated samples, it still correctly defines the support over the sub-spaces:

$$\text{supp}_S(h) = \left\{ s \in \{s_i\}_{i=0}^{N(ha)} \right\} \quad (9)$$

$$\text{supp}_Z(ha) = \left\{ z \in \{z_i\}_{i=0}^{N(ha)} \right\} \quad (10)$$

additional definitions:

$$S = S^{\text{in}} \cup S^{\text{out}} \quad s.t. \quad S^{\text{in}}(h) = \text{supp}_S(h) \quad (11)$$

$$Z = Z^{\text{in}} \cup Z^{\text{out}} \quad s.t. \quad Z^{\text{in}}(ha) = \text{supp}_Z(ha) \quad (12)$$

5 Robust Value

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the nominal value function assumes that the nominal models coincide with the true environment. this assumption might be too strong, since the models may be estimated from data, misspecified, or subject to perturbations. the robust value function tries to provide a meaning full value for a policy when the nominal models are inaccurate, with some error radius. we want to evaluate how well does a policy π performs under the worst plausible model. So instead of evaluating π under one fixed model, we evaluate it over an uncertainty set of possible models. the robust value function is defined as:

$$V_t^{\pi, \text{rob}}(h_t) = \inf_{\theta_{t:H-1} \in \Theta_{t:H-1}} V_t^{\pi}(h_t; \theta_{t:H-1}) \quad (13)$$

such that $\theta_k = (P_T^k, P_Z^k)$ is one step admissible model at time k , $\theta_{t:H-1} = (\theta_t, \dots, \theta_{H-1})$ is one sequence of admissible models over the horizon H , and $\Theta_{t:H-1} = \{\text{set of all admissible model sequences } \{(P_T^k, P_Z^k)_{k=t}^{H-1}\}\}$. this formulation allows non-stationary models, stationary models are a special case such that $\theta_t = \theta_{t+1} = \dots = \theta_{H-1}$ and we get:

$$V_t^{\pi, \text{rob}}(h_t) = \inf_{\theta \in \Theta} V_t^{\pi}(h_t; \theta) \quad (14)$$

6 Rectangularity

motivation: in robust value definition in 13, the uncertainty set $\Theta_{t:H-1}$ may contain arbitrary dependencies between model choices at different time steps, histories, state-action pairs, or observation states. that is clean and does not requires simplifying assumptions, but it is also **computationally very heavy/difficult**. The **adversary's choice is a full model sequence**, and the feasible choices at one time may be coupled with choices at other times. This prevents local dynamic-programming-style decompositions. rectangularity is used to make the robust problem decomposable and allows us to efficiently compute the the robust value function using dynamic-programming-style.

rectangularity $\Rightarrow$ local robust Bellman/backups $\Rightarrow$ tractable online tree computation

Rectangularity means the admissible uncertainty set can be broken down into local uncertainty sets, meaning the the set of all admissible model sequences can be **extended (larger)** into:

$$\Theta_{t:H-1}^{\text{rect}} = \Theta_t \times \Theta_{t+1} \cdots \times \Theta_{H-1} \quad (15)$$

This says that nature can choose each one-step model θ_k independently from its local admissible set Θ_k . the rectangularity is in 2 levels, across time and across models (transition and observation rows may be chosen independently from their local uncertainty sets)

Rectangularity allows the worst-case model to vary across time steps, subject to local feasibility constraints. This represents an adversary that may adapt its model choice locally rather than committing to a single global model.

important to note that the cost of assuming rectangularity leads to a **more conservative** value than the stationary robust value. since $\Theta_{t:H-1} \subseteq \Theta_{t:H-1}^{rect}$, we get:

$$V_{rect}^{rob} = \inf_{\theta_{t:H-1} \in \Theta_{t:H-1}^{rect}} V_t^\pi(h_t; \theta_{t:H-1}) \leq \inf_{\theta_{t:H-1} \in \Theta_{t:H-1}} V_t^\pi(h_t; \theta_{t:H-1}) = V^{rob} \quad (16)$$

rewriting 13 recursively and using rectangularity we get the following form:

$$V_{t,rect}^{\pi,rob}(h_t) = \inf_{\theta_t \in \Theta(h_t)} \mathbb{E}^{\pi, \theta_t} \left[r(s_t, a_t) + V_{t+1,rect}^{\pi,rob}(h_{t+1}) \mid h_t \right]. \quad (17)$$

as seen in 17 the *inf* is only over the current time step, and the minimization of the next time step is computed inside $V_{t+1,rect}^{\pi,rob}(h_{t+1})$, which gives as the dynamic-programming decomposing that allows us to use the bellman backups.

7 Projected Models

should i start by motivation for the projected models here?? given the definitions above of the uncertainty set, we can now discuss the projected models. the projected models $\hat{P}_T, \hat{P}_Z$ are sub-probability vectors and do not correspond to a valid distribution. formally we define them as:

$$\hat{P}_T(\cdot | s, a; h) = \begin{cases} P_T^{\text{in}}(s' | s, a), & s' \in S^{\text{in}}(h), \\ 0, & \text{otherwise.} \end{cases} \quad (18)$$

$$\hat{P}_Z(\cdot | s'; ha) = \begin{cases} P_Z^{\text{in}}(z | s'), & s' \in S^{\text{in}}(h) \wedge z \in Z^{\text{in}}(ha), \\ 0, & \text{otherwise.} \end{cases} \quad (19)$$

important to note that we do not assume anything on how the sub-spaces $S^{\text{in}}, Z^{\text{in}}$ were chosen, for any sub-space the definition above following theory holds. as opposed to the projected models, the belief on the support can be defined differently:

$$\hat{b}(\cdot) = \begin{cases} b(s), & s \in S^{\text{in}}(h), \\ 0, & \text{otherwise.} \end{cases} \quad (20)$$

we can define the simplified belief as "projected belief", instead of the full belief masked. but that will account in higher bounds and double counting, need to discuss with vadim...

8 Simplified Robust Value Function

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motivation: the full robust value is computationally infeasible because it sums over all state-observation trajectories and optimizes over full uncertainty sets. In online planning, the planner only has access to a sampled subset of states and observations at each history/action node. Therefore, the planner evaluates a projected problem on this sampled support. The missing part of the state-observation space is not ignored for free; it becomes part of the approximation error bounded later.

At each history node h and action node ha , the simplified planner restricts the transition and observation sums to the sampled supports $S^{\text{in}}(h)$ and $Z^{\text{in}}(ha)$. let us define the set of valid trajectories, meaning the set of trajectories that the online planner can plan:

$$\mathcal{T}_{t:H-1}^{\text{in}}(h_t) = S^{\text{in}}(h_t) \times \mathcal{A}_t \times Z^{\text{in}}(h_t a_t) \cdots S^{\text{in}}(h_{H-1}) \times \mathcal{A}_{H-1} \times Z^{\text{in}}(h_{H-1} a_{H-1}) \quad (21)$$

such that a single trajectory sample is of the form $\tau_{t:H-1}^{\text{in}} = \{s_t, a_t, s_{t+1}, z_{t+1}, \dots, a_{H-2}, z_{H-1}, s_{H-1}\} \in \mathcal{T}_{t:H-1}^{\text{in}}$. important nuance in here is that the support is a function of the history, so in 21 the product between the sub-spaces is not constant, meaning that if at $t = 3$ we sampled $a' \in \mathcal{A}_3$ then $Z^{\text{in}}(h_3 a') \neq Z^{\text{in}}(h_3 a'')$ since the action-node $h_3 a' \neq h_3 a''$. so the **set** of valid trajectories is the set of trajectories represented by the sampled planning tree.

alternative definition: from a fixed root history h_t . A trajectory belongs to $\mathcal{T}_{t:j}^{\text{in}}(h_t)$ if, for every $k = t, \dots, j-1$, $s_k \in S^{\text{in}}(h_k)$ the action a_k is an admissible action at h_k , $z_k \in Z^{\text{in}}(h_k a_k)$, the next history is generated by $h_{k+1} = h_k, a_k, z_{k+1}$, and the next state satisfies $s_{k+1} \in S^{\text{in}}(h_{k+1})$

using 21 and 3, given a policy π we can define the simplified value function as:

$$\hat{V}_t^\pi(h_t) = \sum_{j=t}^{t+H} \sum_{z_{t+1:j} \in Z_{t+1:j}^{\text{in}}} \sum_{s_{t:j} \in S_{t:j}^{\text{in}}} \prod_{k=t}^{j-1} P_T^{\text{in}}(s_{k+1} | \pi_k, s_k) P_Z^{\text{in}}(z_{k+1} | s_{k+1}) b(s_t | h_t) r(s_j, a_j) = \quad (22)$$

$$\sum_{j=t}^{t+H} \sum_{\tau_{t:j}^{\text{in}} \in \mathcal{T}_{t:j}^{\text{in}}} \hat{\mu}(\tau_{t:j}) r(s_j, a_j) \quad (23)$$

$$\text{s.t.} \quad \hat{\mu}(\tau_{t:j}) = b(s_t) \prod_{k=t}^{j-1} P_T^{\text{in}}(s_{k+1} | \pi_k, s_k) P_Z^{\text{in}}(z_{k+1} | s_{k+1}) \quad (24)$$

$\hat{\mu}(\tau_{t:j})$ is a short writing for convenience which represents the probability mass for receiving the reward $r(s_j, a_j)$. the simplified value is not a different reward objective; it is the same expected reward, but evaluated only over trajectories represented in the sampled support.

similarly to the definition of the robust value function 13, the simplified robust value is defined as the worst case models over the **projected uncertainty set** given a policy π :

$$\hat{V}_t^{\pi, \text{rob}}(h_t) = \inf_{\hat{\theta}_{t:H-1} \in \hat{\Theta}_{t:H-1}} \hat{V}_t^\pi(h_t; \hat{\theta}_{t:H-1}) \quad (25)$$

$$\text{s.t.} \quad \hat{\theta}_k = (P_{T,k}^{\text{in}} \in \hat{\mathcal{P}}_T, P_{Z,k}^{\text{in}} \in \hat{\mathcal{P}}_Z) \quad (26)$$

9 deterministic Guarantees given a policy π

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in this section we want to analyze the performance guarantees of the simplified robust value function. the following theorem provides **deterministic guarantees** of the transition, observation and belief simplified robust value 25 with respect to the theoretical robust value 13:

Theorem 1. let h_t be the history at time t , H the horizon of the POMDP/MDP. $V_t^{\pi, \text{rob}}(h_t)$ is the theoretical robust value function that is achieved by following the policy π (deterministic/stochastic), and $\hat{V}_t^{\pi, \text{rob}}(h_t)$ is the simplified robust value function 25 following the same policy π . for any such policy π the difference between the theoretical and the simplified value functions is bounded deterministically by:

$$|V_t^{\pi, \text{rob}}(h_t) - \widehat{V}_t^{\pi, \text{rob}}(h_t)| \leq R_{\max} \sum_{j=t}^{t+H} \left(1 - \inf_{\theta_{t:j-1} \in \Theta_{t:j-1}} \sum_{\tau_{t:j}^{i_n} \in \mathcal{T}_{t:j}^{i_n}} \widehat{\mu}(\tau_{t:j}) \right) \quad (27)$$

we use the abbreviation $\widehat{\mu}$ defined in 24 for simplicity. 2 important properties of the bound are, first the bound term contains dependency only on sampled trajectories within the support space which makes it an anytime certification of the policy quality, as it is easily computed during planning and does not add additional overhead. secondly the term $\widehat{\mu} \leq 1$ since it is a product of probability vectors, meaning that as the $(S^{\text{in}} \times Z^{\text{in}}) \rightarrow (S \times Z)$ the bound diminishes:

$$\sum_{\tau_{t:j}^{i_n} \in \mathcal{T}_{t:j}^{i_n}} \widehat{\mu}(\tau_{t:j}) \xrightarrow{S^{\text{in}} \rightarrow S, Z^{\text{in}} \rightarrow Z} 1 \quad (28)$$

this leads to a convergence towards the theoretical robust value as the sampled support increases.

Lemma 1. for a given policy π as the support size $(S^{\text{in}} \times Z^{\text{in}})$ increases the simplified robust value function converges to the theoretical value function:

$$\widehat{V}_t^{\pi, \text{rob}} \xrightarrow{S^{\text{in}} \rightarrow S, Z^{\text{in}} \rightarrow Z} V_t^{\pi, \text{rob}} \quad (29)$$

10 Optimality Guarantees

1. Theorem 2. optimal robust value function is bounded by UDB,
2. exploration with the UDB
3. Lemma 2. using exploration with UDB -> converging to optimal robust value function

11 Methods

12 Complexity

13 Experiments

14 Supplementary

in this section we want to explicitly derive the value function, that is used for the bounds. considering horizon H , the return at time step t is defined as:

$$G_t^\pi = \sum_{j=t}^{t+H} r(s_j, a_j), \quad a_j \sim \pi_j(\cdot | h_j) \quad (30)$$

and the value function is defined as:

$$V_t^\pi(h_t) = \mathbb{E}^\pi[G_t | h_t, b_t] = \mathbb{E}^\pi \left[r(s_t, a_t) + \sum_{k=t+1}^{t+H} r(s_k, a_k) | h_t, b_t \right] = \quad (31)$$

$$\mathbb{E}^\pi[r(s_t, a_t) | h_t, b_t] + \mathbb{E}^\pi \left[\sum_{k=t+1}^{t+H} r(s_k, a_k) | h_t, b_t \right] = \quad (32)$$

$$\mathbb{E}_{s_t \sim b(\cdot | h_t)} [r(s_t, a_t)] + \mathbb{E}_{h_{t+1:t+H} \sim p(\cdot | h_t, a_t, s_t)} [G_{t+1} | h_t, b_t] \quad (33)$$

we first show how to define the history distribution $p(\cdot|h_t, a_t, s_t)$ with the models: [should condition on the policy sequence?]

$$p(h_{t+1:t+H}|h_t, a_t, s_t) = p(a_{t+1:t+H}, z_{t+1:t+H}|a_{0:t}, z_{1:t}, s_t) \quad \underbrace{\hspace{10em}}_{\text{marginalize over states}} \quad (34)$$

$$\sum_{s_{t+1:t+H}} p(a_{t+1:t+H}, z_{t+1:t+H}, s_{t+1:t+H}|a_{0:t}, z_{1:t}, s_t) \quad \underbrace{\hspace{10em}}_{\text{chain rule (states)}} \quad (35)$$

$$\sum_{s_{t+1:t+H}} p(a_{t+1:t+H}, z_{t+1:t+H}|a_{0:t}, z_{1:t}, s_{t:t+H}) \cdot p(s_{t+1:t+H}|a_{0:t}, z_{1:t}, s_t) \quad \underbrace{\hspace{10em}}_{\text{chain rule (observations)}} \quad (36)$$

$$\sum_{s_{t+1:t+H}} p(a_{t+1:t+H}|a_{0:t}, z_{1:t+H}, s_{t:t+H}) \cdot p(s_{t+1:t+H}|a_{0:t}, z_{1:t}, s_t) \cdot p(z_{t+1:t+H}|a_{0:t}, z_{1:t}, s_{t:t+H}) \quad \underbrace{\hspace{10em}}_{\text{markovian}} \quad (37)$$

$$\sum_{s_{t+1:t+H}} \prod_{k=t}^{t+H-1} p(a_{k+1}|h_{k+1}) \cdot p(s_{k+1}|a_k, s_k) \cdot p(z_{k+1}|s_{k+1}) \quad \underbrace{\hspace{10em}}_{\text{writing with models notations}} \quad (38)$$

$$\sum_{s_{t+1:t+H}} \prod_{k=t}^{t+H-1} \pi(a_{k+1}|h_{k+1}) \cdot P_T(s_{k+1}|a_k, s_k) \cdot P_Z(z_{k+1}|s_{k+1}) \quad (39)$$

now we want to plug (39) in (33):[policy should be time-indexed (since it's finite horizon)]

$$\begin{aligned} V_t^\pi(h_t) = & \underbrace{\sum_{a_t} \pi(a_t | h_t)}_{\substack{\text{only if} \\ \text{stochastic}}} \left[\sum_{s_t \in S} b(s_t | h_t) \left[r(s_t, a_t) + \sum_{a_{t+1:t+H}} \sum_{z_{t+1:t+H}} \sum_{s_{t+1:t+H}} \right. \right. \\ & \left. \prod_{k=t}^{t+H-1} \underbrace{\pi(a_{k+1} | h_{k+1})}_{\substack{\text{only if} \\ \text{stochastic}}} P_T(s_{k+1} | a_k, s_k) P_Z(z_{k+1} | s_{k+1}) \left[\sum_{j=t+1}^{t+H} r(s_j, a_j) \right] \right] \end{aligned} \quad (40)$$

we consider a deterministic policy (the tree policy), and therefore we can remove the policy out of the equation: [(i) for deterministic policies, should replace a_j with $\pi_j(h_j)$; (ii) if we later consider MCTS, it calculates a stochastic policy]

$$\begin{aligned} V_t^\pi(h_t) = & \sum_{s_t \in S} b(s_t | h_t) \left[r(s_t, a_t) + \sum_{z_{t+1:t+H}} \sum_{s_{t+1:t+H}} \prod_{k=t}^{t+H-1} P_T(s_{k+1} | a_k, s_k) P_Z(z_{k+1} | s_{k+1}) \left[\sum_{j=t+1}^{t+H} r(s_j, a_j) \right] \right] \\ = & \sum_{j=t}^{t+H} \sum_{z_{t+1:j}} \sum_{s_{t:j}} \prod_{k=t}^{j-1} P_T(s_{k+1} | \pi_k, s_k) P_Z(z_{k+1} | s_{k+1}) b(s_t | h_t) r(s_j, a_j) \end{aligned} \quad (41)$$

we consider (41) as the theoretical value function. similarly the simplified value function given that we only see a subset (inside support) of the space is define as:

$$\hat{V}_t^\pi(h_t) = \sum_{j=t}^{t+H} \sum_{z_{t+1:j} \in Z_{t+1:j}^{\text{in}}} \sum_{s_{t:j} \in S_{t:j}^{\text{in}}} \prod_{k=t}^{j-1} P_T^{\text{in}}(s_{k+1} | \pi_k, s_k) P_Z^{\text{in}}(z_{k+1} | s_{k+1}) b^{\text{in}}(s_t | h_t) r(s_j, a_j) \quad (42)$$